

Co-Designing an Experience Sampling Method Digital Platform to Investigate Self-Harm Among Young People

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ABSTRACT

Self-harm is particularly prevalent amongst young people, with adverse consequences in terms of later wellbeing, vocational outcomes, and increased risk of suicide. Although many studies have explored self-harm, it remains difficult to predict and prevent, in part because the mental states that typically precede self-harm are highly variable and triggers can be unclear. The Experience Sampling Method (ESM), where participants are asked to record personal data many times during the day, allows researchers to capture these variable mental states and their relation to self-harming behaviour. We conducted a series of co-design workshops involving young people with lived experience of self-harm and researchers with a special interest in ESM to identify the requirements of both groups for an ESM digital platform for investigating self-harm. We describe their key requirements, some of which are conflicting, and suggest ways that these could be addressed to develop an effective ESM platform.

CCS CONCEPTS

• **B7; Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Self-Harm, Experience Sampling Method, Co-Design, Young People

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1 INTRODUCTION

Self-harm, defined here as any act of self-injury or poisoning, irrespective of intent, is particularly prevalent among young people (10–24 years), with prevalence rates being estimated at 12–31% in western countries [1]. Self-harm among young people has several adverse outcomes, including unemployment and poor mental health in adulthood, and is strongly associated with subsequent suicidal behaviour [1]–[3]. Given the long-term consequences of self-harming behaviours there is a need to understand the causes of self-harm in young people. However, self-harm remains difficult to predict and prevent [4].

Data regarding self-harm is often collected retrospectively, which is likely to be heavily susceptible to biases, including recall bias and social desirability bias [5], [6]. Furthermore, although cross-sectional (all data collected at a single timepoint) and longitudinal studies may help to understand distal risk factors, such as female gender, they do not help to elucidate immediate contextual factors, such as triggers for self-injurious thoughts or what factors may predict transition from self-injurious thoughts to behaviours [7], [8].

The Experience Sampling Method (ESM) is a group of techniques involving frequent measurements of participants in their daily lives. These techniques typically consist of participants completing repeated, short questionnaires to provide in-the-moment reports [9]. These questionnaires can be delivered through different means, including electronically and via paper-based surveys [10]. A review of studies using ESM with individuals who have a history of self-harm demonstrated that this methodology provides valuable new insights into why people self-harm [5]. For example, a previous cross-sectional study with patients seeking treatment for Borderline Personality Disorder (BPD) revealed that 'emotion regulation' was the most prevalent motivation for engaging in self-harm and 'self-punishment' the second most prevalent [11]; however, although Andrewes et al. (2017) ESM study also found that emotion regulation was the most commonly reported motivation, only 9% of participants cited self-punishment as a motivation for self-harming [12].

The widespread availability of mobile technologies may help facilitate ESM and thus advance the understanding of self-harm as well as laying the foundations for future treatments and interventions [13]. ESM can present challenges in terms of ethical concerns, research design, and feasibility. ESM studies allow greater levels of

flexibility in study design resulting in a greater number of choices for researchers to make. For example, the duration of ESM studies can vary from days, to weeks and even months, and the frequency of questionnaires can also range from 1 to more than 50 per day [14]. Identifying the appropriate study duration and questionnaire frequency is critical, as the burden associated with having to respond to too many questionnaires can lead to high levels of participant drop out, which impacts findings [15]. Furthermore, research suggests that *how* questions are presented to users matters, yet the research into the effects of different design choices in ESM studies is limited [14].

The importance of involving young people with lived experience as active partners in research is increasingly recognised, with many policy makers and funding bodies now mandating user involvement [16]. Bailey et al (2020) describes benefits that result from youth partnerships, including greater relevance of research questions, improved recruitment and retention rates, better long-term outcomes and ensuring that interventions and methodologies are youth-friendly. Barriers to partnerships in research have also been identified, related to a lack of knowledge or confidence on the part of the researchers, as well as resource limitations such as funding, training, and capacity to provide adequate support for participants [17]. These barriers may be exacerbated by the sensitive nature of topics such as self-harm.

Our goal is to develop a set of design guidelines for building an effective ESM platform to investigate self-harm in young people. We therefore conducted a series of co-design workshops, with young people with lived experience of self-harm and researchers with an interest in ESM. This paper reports our initial findings related to the following research questions:

- What design requirements do (i) young people with lived experience of self-harm and (ii) ESM researchers have for an ESM platform?
- What are the contrasts between what the researchers and young people require, and can these be overcome through design?

2 METHOD

The research presented in this paper received ethical approval from the University of Bristol research ethics committee.

2.1 Participants

Fourteen young people with lived experience of self-harm, with ages ranging from 15-23 years, (median; 17; eight female) were recruited via social media platforms and charity organisations. All participants provided written informed consent to take part in the study and for participants under the age of 18 written parental assent was also required. Participants were provided with the Participant Information Sheet when they expressed interest in participating in the study. Participants received a £10 shopping voucher for each session attended. Upon completion of all three sessions participants received an additional £10 voucher.

Recruitment and retention are particularly challenging within this target group. A key challenge when considering the study design was the continually changing situation of COVID-19 social distancing and lockdown measures. Restrictions meant we initially focused on recruiting people online. Social media has the

potential to make research more publicly accessible, while enabling researchers to access communities and participant groups which were previously inaccessible. We used Instagram, Facebook and Twitter for recruitment, but with limited effect. Many individuals who completed our screening survey direct from social media did not engage with further steps of the study. Different charity organisations were then contacted to see if they could help with recruitment using their online platforms and email lists. Our most successful avenue of recruitment was through engaging with MQ Participate, a mental health research charity.

Five researchers, ranging from PhD Students to senior researchers (two female), were recruited through an ESM working group at the University of Bristol, by emailing the mailing list for the group, comprising institutional email addresses. Researchers were not compensated for attending the workshop.

2.2 Workshops with Young People

Thirteen semi-structured workshops were conducted with the fourteen young people organised into four groups. Each participant was invited to attend a sequence of three workshops.

Due to COVID-19 lockdown measures and restrictions, the co-design workshop sessions were conducted remotely using University of Bristol approved videoconferencing software. Although participants were encouraged to have their cameras on this was not a requirement. The sessions were audio-recorded and transcribed. Co-designing with young people virtually was challenging as it was harder to build a rapport with participants. Nearly all the young people kept their cameras off throughout all the sessions, as well as often muting their microphones. This impacted the flow of the sessions as well as hiding many social cues associated with group discussions and design activities. A Child and Adolescent Psychiatrist (RM) helped facilitate the workshops with the young people and was also present to help if any participant unexpectedly dropped out of the session. No participants dropped out of sessions or reported needing support as a result of participating in the study.

The first workshop session aimed to familiarise participants with the facilitators as well as key topics which would be discussed. It explored young people's use of mobile technologies as well as their thoughts on ESM and how this approach could be used in the research of self-harm among young people.

The second session involved presenting participants with a range of fictional scenarios relating to the research of self-harm in the community. Scenarios depicted an individual who receives a notification on a smartphone prompting them to input responses on an EMA app. The scenarios differed in terms of time of prompt; individual location at time of prompt; activity at time of prompt; phone activity at time of prompt. Contextual features of each scenario were informed by previous research [18].

The final session involved presenting participants with a visual prototype of an ESM platform. This prototype was informed by the data gathered in the previous sessions and developed in Adobe XD. Participants were shown a demonstration of the platform starting with onboarding screens, security measures, settings features, and question formats. While the prototype was shown to participants, they were asked to discuss each feature of the application prototype. The group discussion was initially focused on what young people

liked and disliked about the prototype, including the language and design features, and then focused on their views of using ESM to research self-harm in young people.

2.3 Workshops with ESM Researchers

We recruited five academic researchers with an interest in ESM and conducted two workshops to gather their requirements for an ESM platform to investigate self-harm. We also elicited their reactions to the prototype developed to meet the requirements of young people with lived experience of self-harm. The sessions were semi-structured and to aid discussion, researchers were presented with two prototypes: a low fidelity dashboard for an ESM platform; and the ESM app prototype presented to young people in their final workshops.

The transcripts from all of the workshops were analysed following an iterative inductive thematic analysis [19].

3 FINDINGS

In this section we first describe the themes that emerged from the thematic analysis of the workshops with young people. We then report a key theme from the researcher workshops and focus on their responses to the preferences of young people.

3.1 Privacy and Security

Young people raised many concerns about issues relating to the privacy and security of using an ESM platform. Young people were not concerned about privacy and security in general, or that apps on their smartphones could track their location. However, they did worry about this in relation to self-harming: *“But when it’s talking about some quite personal stuff you get a bit worried the phone knows where you are, so who else has access to this? And I think it’s just because the topic around self-harm is quite a difficult subject for many.”* (YP4).

Participants universally agreed that the ESM platform should be discreet and ensure the confidentiality of the young people being studied. However, one individual did question if this privacy could contribute to the stigma surrounding self-harm and fed into a narrative of hiding the behaviour and not talking about it; *“I can see how people want it private so . . . [the app] locks and no one can access, but does it make you a bit secretive? Does it make self-harm even more, ‘Oh, let’s not talk about this, let’s hide it behind something else as well.’”* (YP4).

All participants wanted features of the app, including prompts or notifications, not to indicate that the app was to do with self-harm. *“I think it would have to be like discreet, in [...] like the design of the app and because yeah, I think otherwise it could be a bit [...] invasive”* (YP2). Participants thought having a generic home screen when the ESM app was opened was an effective way of providing confidentiality and preventing others knowing what the app was about: *“I think a home screen might be good because that could again make it a bit more discreet.”* (YP1). Participants wanted the app to have privacy features, such as only being able to open the app with a passcode or face or fingerprint recognition.

The context of use influenced how comfortable participants felt about using an ESM platform. Young people expressed a clear preference for using the platform when they were alone, and had more

confidentiality concerns about using the platform when they were with friends or family or in a public situation where someone could look over their shoulder and see the app *“I think the right place that I can use the app is when maybe I’m free and away from the public because some of it needs to be kept private, you know some people are nosey”* (YP6). All participants thought it was important that other people not know what the ESM platform was about, *“Yeah, I wouldn’t want like if I was with people, and I wouldn’t want anyone to see it and know what it was about”* (YP1).

3.2 Impact on daily life

Young people were concerned about the impact of participating in ESM research, in particular, they were concerned about the commitment that would be required and how it would fit into their everyday lives.

Young people discussed how many questionnaires they would be able to answer each day. They also thought that keeping questionnaires short was important to ensure engagement: *“they have to be short answers for people to keep engaged. If you’re having to write paragraphs then, people won’t want to do it”* (YP2). Participants stated that a mixture of questions asked at different times of day would make them think more about their answers and prevent them from giving automatic, unconsidered responses.

The issue of ESM being ‘annoying’ was raised repeatedly and seen as a key consideration in the design of ESM studies. Young people were concerned about being bombarded with tasks and repeated questions. Participants wondered how much commitment this would be in their everyday lives, and how the potential annoyance could be mitigated, *“it sounds like quite a big commitment to have to like take every so often out of your day to answer loads of questions [...] I don’t know if many people would actually want to do that”* (YP5).

Some young people thought that if the ESM app became too annoying, they would ‘swipe off’ the questions to avoid inputting a response. Young people suggested ways that the ESM app could be less annoying, for example, that the questions could vary each time they entered responses *“I just think the repetition would get a bit annoying, like constantly seeing the same question.”* (YP3).

Participants highlighted the importance of timing questions to limit the inconvenience, *“[if I] just want to express myself but [sometimes] due to the location I like just find it so hideous. So, I think sometimes it affects so much.”* (YP12). One of the scenarios presented in session two depicted a young person travelling home on a bus and this prompted discussion about the potential benefit of having the time to input a response in this situation, and the potential downside of the presence of other people. Young people generally thought that times when they were with others, such as friends or family, would not be a good time to input responses: *“I think it depends if I’m with other people or not so, if I was on my own, I’d be more likely to respond but if I was with my friends then I would probably just ignore it”* (YP1).

Levels of distress were also discussed as a factor that would impact whether they wanted to respond to questionnaires or not, with some participants reporting that they believed when people were distressed it was unlikely that they would want to input responses. However, one participant stated that if they had had a

bad day, they would be more likely to engage with the app and respond: “*it depends because say I was having a bad day then [...] I’ll be more willing to like write how I’m feeling down and answer questions*” (YP5).

Additionally, participants highlighted the need for sensitively designed questionnaires, both in terms of the wording of individual questions and the ordering of sequences of questions. In particular, one participant emphasised that they thought it was best to initially ask whether a person had had any thoughts about self-harm and only if they answered in the positive to follow up with a question asking whether they had self-harmed: “*I think it’s better to ask first if you’ve had any thoughts [of self-harm] because if you just get asked, ‘Have you [self-harmed]?’ and like you may have wanted to, but you haven’t and it kind of invalidates like that you’ve had the thoughts [but not acted on them]*” (YP1). More generally, when discussing self-harm young people stressed the importance of taking a whole person approach and not looking at self-harm as an isolated incident but rather how it is integrated within their lives “*I think the obvious question [that] is going to be in the app is, ‘have you self-harmed?’ ... but I think it’s important to get sort of a whole person approach and get to know the whole picture rather than in just one question*” (YP8). Asking questions about general wellbeing and looking at how self-harm may co-exist with other mental health issues was preferable for most participants.

3.3 App Design

All of the young people expressed a requirement that an ESM app must be easy to use and fit into their current routine. Smartphones were the most used mobile technology, “*I don’t really use like anything other than my phone*” (YP14). Because of this, young people considered smartphones to be the ideal way to deliver ESM as it would already be integrated into their everyday lives “*because it’s kind of something that everybody has and that like everybody uses and it’s not like integrating like something else into like your life or like your routine*” (YP2). Young people wanted the design to be simple, clear, and easy to use, and not overcomplicated or having unnecessary features. Participants commonly drew comparisons with apps designs they were familiar with when discussing desirable features for an ESM app: “*I think all of the big apps are easy to use*” (YP4). Social media platforms, such as Instagram and TikTok, were considered well designed and easy to navigate “*I think like, just keeping it as simple and easy to use as possible like I think that’s why TikTok is so popular because it’s literally just like a home page and you scroll and that’s essentially it*” (YP5).

Of key importance to young people was the use of a friendly design that used informal language. Participants wanted the app to be informal, rather than clinical or echo a healthcare professional in any way. This included features in both design and language “*nothing too like formal almost like it’s clinical in the sense that you’re making them fill out a questionnaire*” (YP5). Additional functionality within the ESM platform was considered as a way to both improve the app and engagement with young people. Commonly cited functions included being able to reflect on their previous responses, having an area for free text input within each questionnaire and providing support: “*You could have like a history or just like a calendar that you could click into to see your previous*

responses maybe” (YP2); “*you could at the bottom of the app or something would be, do you want the space to express anymore of your feelings? And they can just type away and sort of do that.*” (YP4); “*... it should also have a support system.*” (YP6).

Whilst some young people thought having support integrated into the ESM platform may be beneficial, others discussed the need to manage users’ expectations of the app and the researchers’ role in monitoring participants responses “*you’re not a crisis line, it’s not your job to be monitoring for crises [...] If people think that it [their responses] is monitored and they are going to get support that way [...] it’s not fair on the young person or the researchers to have to pick it up [...] so I think it would be best for everyone to have no access [to the participants responses] until it’s [the study] all over.*” (YP4). This discussion about whether the app should provide support reflected the challenge in describing the purpose of the app to young people, that is, as a way of conducting research into self-harm. However, young people often discussed the app as though its purpose was to be an intervention rather than a research tool.

Young people thought it was important that they could personalise the app and feel it was a space in which they could express themselves “*it’s tailored to you and it feels a bit more personal with like you can answer questions around what you have been doing*” (YP7). More generally, young people thought it important that they had control over the timings of the questions as well as other aspects of the app.

Young people wanted to be able to control how many times per day they are prompted to input responses, which hours of the day they can be prompted, how and when they get notified that there is a questionnaire for them to fill in: “*for the app it’d be quite useful if you could set like times when you can be asked the questions. So like if you could set the app saying that you can’t answer questions between 8 and 4 or something like that because you’re at school*” (YP4), “*I think depending on each and every person’s flexibility they can choose to input their responses depending on the times are day they are best comfortable with and they best feel that it’s consistent with them*” (YP8). Young people considered this level of participant control important as everyone has different schedules and preferences as well as different variability in self-harm related thoughts and behaviours “*I think depending on each and every person’s flexibility they can choose to input their responses depending on the times of day they are best comfortable with, and they best feel that it’s consistent with them*” (YP6). Through being able to change the questionnaire schedule, that is, the time of day they are sent, and frequency participants felt they could integrate using the app more easily into their everyday lives. “*It puts the young person fully in control of when they can and when they don’t want to answer*” (YP10).

3.4 Researchers’ Responses

Researchers also wanted control and flexibility in an ESM platform so that they could use it for a range of studies. In particular, they wanted to be able specify the content of the questionnaires, specify different sorts of question and set the schedule and frequency of questionnaires: “*The most important thing is flexibility, especially if you are wanting it to be used by different researchers for different projects and things so ... like having the ability to choose a number of*

times, having the ability to either choose those times or maybe have them at random times, having the ability to choose what times of day like you could say random times between 8:00 AM and 8:00 PM for example” (R4). Other key features that researchers wanted in an ESM platform included summary statistics (for example, compliance and completion rates) flexible question formats, and the ability to use it for varied study protocols. Researchers also stressed the need to be able to adapt and change the user interface and manage the experience users would have with the app. These changes were discussed in line with varying research aims, study protocols and different user groups, *“I think it’s useful to have that sort of flexibility”* (R1), *“so if you could incorporate those kind of [question format] options, I think that would be really, really useful to researchers”* (R3).

Researchers could understand why young people wanted to be able to specify the schedule and frequency of questionnaires but were concerned about the impact this could have on their research: *“I can see it being a nice thing to have for the participant, but for the researcher it could be a bit of a nightmare trying to compare the data between participants”* (R1). One researcher thought that it could be *“a disaster for researchers to have participants doing different things”* (R4). There were concerns that making questionnaire notifications discreet could impact the effectiveness of an ESM study, *“because the whole point is that you’re trying to nudge them into actually getting a response to a question. So, if you remove the notifications, I think that would be very difficult”* (R3).

Researchers were also concerned about giving participants access to their previously recorded responses: *“As a researcher, it’s tricky because you might not want participants to be able to see that [self-reported mood] . . . because you don’t want it to necessarily influence things”* (R4). Additionally, some researchers were sceptical about the research value of the app providing support for users: *“the more likely we are to benefit the mental health of the participants, the less us as researchers are going to learn that we can take forward to new participants. Because if everybody gets better, we’re not going to understand anything about the interplay between self-harm and whatever we’re interested in because we’ve just spoilt it by making everybody better”* (R5).

4 DISCUSSION

The co-design workshops provided insights into the requirements for an ESM platform of both young people with lived experience of self-harm and researchers. One of the benefits of ESM for researchers is that they can measure highly variable phenomena, within participants’ everyday lives. However, this can be invasive and result in high levels of burden for participants, while also raising issues of privacy, security, and control. These issues can be heightened when researching a stigmatised phenomenon like self-harm. As such, we must consider the needs of the participants not only to reduce poor quality data or high levels of drop out but, perhaps more importantly, because we are asking them to make a significant commitment to respond to multiple questionnaires during their daily lives.

An ethical consideration arose in managing young peoples’ expectations of an ESM app. As raised by a young person, participants may expect someone to be monitoring responses in real time and then contact them if a high-risk situation arises. It would typically

not be feasible for researchers to regularly monitor participants’ responses due to constraints on both time and resources. Additionally, if the ESM platform was to provide such levels of support to participants this could impact study outcomes as it would become interventional rather than observational, which was a key concern raised by researchers.

Many features that the young people identified were compatible with the requirements of researchers, for example making the ESM platform discreet to ensure that it is private and secure as well as being simple in design and easy to use. An issue identified by young people was the importance of the language used in questions and the sequence in which questions are presented. For example, it was important for young people that they were first asked if they had thoughts of self-harm, before asking if they had engaged in any self-harm related behaviours. This is because they want recognition of the fact that they might not have acted on self-injurious thoughts.

Whilst many of requirements identified by young people were compatible with the requirements of researchers, conflicting requirements were also identified. Although researchers could empathise with young people wanting to control where and when they responded to questions, researchers were also concerned about giving participants too much control which could reduce the effectiveness of ESM as a research tool. For young people having the ability to adapt how they engaged with the app was considered beneficial to enhance compliance and engagement. Young people saw this control as a means to help them integrate the ESM technology within their daily lives and reduce the associated burden of using the app. However, for researchers, handing over this level of control was considered to put the benefits associated with ESM, in terms of increased validity and accuracy, at risk. There appears to be a trade off in control between the two stakeholders, where giving more control to ESM participants can impact the value of the approach for researchers. In contrast, if researchers have sole control over the frequency and timing of questionnaires then this can increase the burden on participants and potentially lead to higher drop out levels and reduced quality of responses.

The ultimate goal in this line of research is the prevention of self-harm among young people. This paper builds upon previous ESM research through providing new information about how technology can be used to better understand self-harm as it happens in young people. Research into the characteristics of self-harm is essential to be able to predict and prevent self-harming behaviour among young people. Our findings have yielded important insights into how HCI researchers can design technologies to understand self-harm and enhance the validity and quality of ESM assessments for each individual.

5 CONCLUSION

Through co-design workshops we have identified requirements of both researchers and young people with lived experience of self-harm who would be using the ESM platform. Although many of the requirements of these two groups are compatible, there is a potential conflict between the needs of researchers and young people regarding who controls the frequency and timing of questionnaires. Our future research will investigate the extent to which the design of the ESM platform could resolve the different requirements

of these two groups. These findings are limited to young people with lived experience of self-harm and cannot be extended to other groups of ESM users. Nevertheless, the findings are of importance as there is a need for further research into self-harm among young people and ESM has the potential to provide valuable insights.

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